

Intramuscular MAPK signaling following high volume and high intensity resistance exercise protocols in trained men

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Abstract

Purpose To examine the mitogen-activated protein kinase (MAPK) family of signaling proteins following typical high volume (HV) and high intensity (HI) lower body resistance exercise protocols in resistance-trained men.

Methods Ten resistance-trained men (24.7 ± 3.4 year; 90.1 ± 11.3 kg; 176.0 ± 4.9 cm) performed each resistance exercise protocol in a random, counterbalanced order. The HV protocol utilized a load of 70 % 1-RM for sets of 10–12 repetitions with a 1-min rest period length between sets and exercises. The HI protocol utilized a load of 90 % 1-RM for sets of 3–5 repetitions with a 3-min rest period length between sets and exercises. Both protocols included six sets of barbell back squats and four sets of bilateral leg press, bilateral hamstring curls, bilateral leg extensions, and seated calf raises. Fine needle muscle biopsies of the vastus lateralis were completed at baseline (BL) and 1-h post exercise (1H).

Results No significant differences over time were noted for phosphorylation of MEK1, ERK1/2, p38, MSK1, ATF2, p53, or c-Jun ($p > 0.05$). No significance between

trial interactions was noted for phosphorylation of MAPK signaling proteins, including MEK1, ERK1/2, p38, JNK, MSK1, ATF2, STAT1, p53, c-Jun, or HSP27 ($p > 0.05$). However, significant time effects were observed for phosphorylation of JNK ($p < 0.01$), HSP27 ($p < 0.01$), and STAT1 ($p = 0.03$). Phosphorylation of JNK, HSP27, and STAT1 was significantly elevated from BL at 1H for both HV and HI.

Conclusions HV and HI lower body resistance exercise protocols appear to elicit similar MAPK activation in resistance-trained men.

Keywords Mitogen-activated protein kinase · Anabolic intramuscular signaling · ERK · p38 · JNK

Abbreviations

1-RM	1-repetition maximum
1H	1-h post exercise
ANOVA	Analysis of variance
ATF2	Activating transcription factor 2
AU	Arbitrary units
BL	Baseline
ERK1/2	Extracellular signal-related kinase 1/2
HI	High intensity
HPL	Human Performance Laboratory
HSP27	Heat shock protein 27
HV	High volume
JNK	c-Jun N-terminal kinase
MAPK	Mitogen-activated protein kinase
MEK1	Mitogen-activated protein kinase 1
MSK1	Mitogen- and stress-activated protein kinase 1
mTORC1	Mammalian target of rapamycin complex 1
p70S6k	p70S6 kinase
RPS6	Ribosomal protein S6
STAT1	Signal transducer and activator of transcription 1

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Introduction

Resistance exercise initiates a multifaceted series of events converting the stimulus of muscle contraction into biochemical responses regulating the rate of protein synthesis (McGlory and Phillips 2015). The mitogen-activated protein kinase (MAPK) family of signaling proteins has been identified as a key signaling pathway involved in the regulation of skeletal muscle mass (Haddad and Adams 2004; Shi et al. 2009). MAPK signaling can be activated under a range of conditions, including growth factor stimulation (e.g., insulin-like growth factor-1) and muscle contraction (Morrison 2012). Proteins specific to the MAPK signaling cascade have the potential to regulate a range of cellular processes involved in cellular growth, including proliferation and differentiation (Widegren et al. 2001). Upon upstream activation, mitogen-activated protein kinase 1 (MEK1) activates extracellular signal-related kinase 1/2 (ERK1/2), which subsequently activates downstream nuclear transcription factors involved in muscle remodeling, including mitogen- and stress-activated protein kinase 1 (MSK1) and signal transducer and activator of transcription 1 (STAT1) (Farooq and Zhou 2004; Guan 1994). The stress-activated protein kinases, c-Jun N-terminal kinase (JNK), and p38 are also members of the MAPK signaling cascade which activate downstream transcription factors, including activating transcription factor 2 (ATF2), p53, and c-Jun (Widegren et al. 2001). In addition, p38 appears to activate heat shock protein 27 (HSP27), which may also promote exercise-induced muscle remodeling (Landry and Huot 1995). Therefore, an upregulation of MAPK signaling proteins may be a potential mechanism, by which resistance exercise augments muscle specific gene expression and hypertrophy (Widegren et al. 2001).

Manipulation of acute program variables of resistance exercise (e.g., intensity, volume, and rest interval) is known to induce differential phenotypic adaptations specific to the nature of the stimulus (Egan and Zierath 2013). While several human studies have demonstrated an upregulation of MAPK signaling proteins following a bout of resistance training (Deldicque et al. 2008; Holm et al. 2010; Hulmi et al. 2012; Karlsson et al. 2004; Williamson et al. 2003), few have investigated MAPK signaling proteins in response to different resistance exercise paradigms. Phosphorylation of ERK1/2 and p38 has been reported to be greater following high intensity [70 % 1-repetition maximum (RM)] compared to very low intensity (16 % 1-RM) resistance exercise (Holm et al. 2010). However, others have found no differential response in the activation of ERK1/2 when comparing higher (85 % 1-RM) and lower (65 % 1-RM) intensities of resistance exercise (Taylor et al. 2012). Furthermore, Hulmi et al. (2012) reported greater

ERK1/2 phosphorylation following a high-volume, moderate intensity leg press protocol (80 % 1-RM, 5 × 10 repetitions, 2-min rest interval) compared to a very low-volume, high-intensity leg press protocol (100 % 1-RM, 15 × 1 repetition, 3-min rest interval). However, phosphorylation of p38 and ERK1/2 was not significantly different following resistance exercise involving one, three, or five sets of 6-RM in the leg press exercise (Terzis et al. 2010). Therefore, the effect of resistance exercise intensity and volume on the regulation of MAPK signaling appears to remain equivocal. Nevertheless, each of these investigations was conducted using previously untrained individuals and may not be generalized to a trained population.

The MAPK signaling cascade appears to be an important hypertrophic signaling pathway in skeletal muscle; however, the effects of different resistance exercise protocols have not been established among resistance-trained individuals. Alterations in exercise intensity and volume influence the mechanical and metabolic stresses that are believed to induce anabolic intramuscular signaling and muscle growth (Gonzalez et al. 2015a). Thus, the purpose of this study was to examine MAPK signaling proteins following typical high volume (HV) and high intensity (HI) lower body resistance exercise protocols in resistance-trained men.

Materials and methods

Participants

Ten resistance-trained men (24.7 ± 3.4 year; 90.1 ± 11.3 kg; 176.0 ± 4.9 cm) were recruited to participate in this randomized, cross-over design research study. Inclusion criteria required participants to be between the ages of 18 and 35 years, have a minimum of 1 year of resistance training experience, have been resistance training 2–3 times per week at the time of enrollment, and have the ability to squat a weight equivalent to their body mass. Participants had 6.7 ± 4.6 years of resistance training experience with an average maximum barbell back squat of 172.7 ± 25.2 kg. All participants were familiar with the exercises and training intensities employed in the study. All participants were free of any physical limitations that may affect performance. In addition, all participants were free of medications as well as performance enhancing drugs and supplements, as determined by a health and activity questionnaire. Following an explanation of all procedures, risks, and benefits, each participant provided his informed consent prior to participation in this study. The research protocol was approved by the New England Institutional Review Board prior to participant enrollment.

Maximal strength testing

Prior to experimental trials, participants reported to the Human Performance Laboratory (HPL) to establish maximal strength (1-RM) on all lifts involved in the exercise protocol. Prior to maximal strength testing, participants performed a standardized warm-up consisting of 5 min on a cycle ergometer against a light resistance, 10 body weight squats, 10 body weight walking lunges, 10 dynamic walking hamstring stretches, and 10 dynamic walking quadriceps stretches. The 1-RM test for the barbell back squat and leg press were performed using methods previously described by Hoffman (2006). Briefly, each participant performed two warm-up sets using a resistance of approximately 40–60 and 60–80 % of his perceived maximum, respectively. For each exercise, 3–4 subsequent trials were performed to determine the 1-RM. A 3–5-min rest period was provided between each trial. For all other exercises, the 1-RM was assessed using a prediction formula based on the number of repetitions performed to fatigue using a given weight (Brzycki 1993). Trials not meeting the range of motion criteria for each exercise or where proper technique was not used were discarded.

Experimental trials

On the morning of each trial, participants reported to the HPL after a 10-h overnight fast and having refrained from all forms of moderate to vigorous exercise for the previous 72 h. Experimental trials were performed in a balanced, randomized order, and each experimental trial was separated by a minimum of 1 week to ensure adequate recovery. Each participant performed experimental trials at the same time of day to avoid diurnal variations.

During each experimental trial, participants performed the standardized warm-up routine described above, followed by a lower body resistance exercise protocol. Table 1

shows the HV and HI resistance exercise protocols. The HV protocol utilized a load of 70 % 1-RM for sets of 10–12 repetitions with a 1-min rest period length between sets and exercises. The HI protocol utilized a load of 90 % 1-RM for sets of 3–5 repetitions with a 3-min rest period length between sets and exercises. Both protocols included six sets of barbell back squats and four sets of bilateral leg press, bilateral hamstring curls, bilateral leg extensions, and seated calf raises. During each protocol, participants were verbally encouraged to complete each set. If the participant was unable to complete the desired number of repetitions, spotters provided assistance until the participant completed the remaining repetitions. Subsequently, the load for the next set was adjusted, so that participants were able to perform the specific number of repetitions for each set.

Following each resistance exercise protocol, participants remained in the laboratory for post-exercise assessments. Fine needle muscle biopsies were completed at baseline (BL) and 1-h post exercise (1H). As heavy resistance exercise is not typically performed in the fasted state, participants were provided a standardized low protein, low carbohydrate breakfast bar (Atkins Nutritionals, Inc., Denver, CO, USA: 7 g protein, 3 g carbohydrate, and 3 g fat) following BL assessments to control for diet. Immediately following resistance exercise, participants were also provided a flavored drink (355 mL, 0 g protein, 2.5 g carbohydrates, 0 g fat) to limit any nutritional impact on the signaling response.

Fine needle muscle biopsy procedure

Fine needle muscle biopsies were performed on the vastus lateralis muscle of the participant's dominant leg using a spring-loaded, reusable instrument with 14-gauge disposable needles and a coaxial introducer (Argon Medical Devices Inc., Plano, TX, USA). Biopsy sampling of a small population of muscle fibers was assumed to be representative of the entire

Table 1 Resistance exercise protocols

Exercise order	High volume protocol (HV)			High intensity protocol (HI)		
	Sets × repetitions	Intensity	Rest time	Sets × repetitions	Intensity	Rest time
1. Barbell back squats	6 × 10–12			6 × 3–5		
2. Bilateral leg press	4 × 10–12			4 × 3–5		
3. Bilateral hamstring curls	4 × 10–12	70 % 1-RM	1 min	4 × 3–5	90 % 1-RM	3 min
4. Bilateral leg extensions	4 × 10–12			4 × 3–5		
5. Seated calf raises	4 × 10–12			4 × 3–5		

The HV protocol utilized a load of 70 % of one repetition maximum (1-RM) for sets of 10–12 repetitions with a 1-min rest period length between sets and exercises. The HI protocol utilized a load of 90 % 1-RM for sets of 3–5 repetitions with a 3-min rest period length between sets and exercises. If the participant was unable to complete the desired number of repetitions, spotters provided assistance until the participant completed the remaining repetitions. Subsequently, the load was adjusted, so that participants were able to perform the specific repetitions for each set

muscle. Following local anesthesia with 2 mL of 1 % lidocaine applied into the subcutaneous tissue, a small incision to the skin was made and an insertion cannula was placed perpendicular to the muscle until the fascia was pierced. The biopsy needle was inserted through the cannula, and a muscle sample was obtained by the activation of a trigger button, which unloaded the spring and activated the needle to collect a muscle sample. Multiple biopsy passes at each time point were made with the cannula in place, thus avoiding repeated skin punctures. Each muscle sample was removed from the biopsy needle using a sterile scalpel and was, subsequently, placed in a cryotube, rapidly frozen in liquid nitrogen, and stored at -80°C . All muscle biopsies were performed by a licensed physician.

Intramuscular anabolic signaling analysis

Tissue samples were thawed and kept on ice for preparation and homogenization. A lysis buffer with protease inhibitor (EMD Millipore, Billerica, MA, USA) was added to each sample at a rate of 500 μL per 10 mg of tissue. Samples were homogenized using a Teflon pestle and sonication (Branson, Danbury, CT, USA). Tissue samples were then placed on a plate shaker (Thermo Fisher Scientific Inc., Waltham, MA, USA) for 10 min at 4°C and, subsequently, centrifuged at $10,000\times g$ for 5 min. The supernatant was aspirated and used for analysis.

Multiplex ELISA was used to quantify the phosphorylation status of proteins specific to the MAPK signaling pathway using MAGPIX[®] (Luminex, Austin, TX, USA) and a multiplex signaling assay kit (EMD Millipore, Billerica, MA, USA) according to manufacturer's guidelines. Multiplex ELISA has been validated (Hwang 2011) and previously used to determine the phosphorylation status of signaling proteins (Gonzalez et al. 2015b, c; Sharma et al. 2012a, b). Samples were analyzed for phosphorylation of MEK1 at Ser 217/221, ERK1/2 at Thr 185/Tyr 187, p38 at Thr 180/Tyr 182, JNK at Thr 183/Tyr 185, STAT1 at Tyr 701, MSK1 at Ser 212, ATF2 at Thr 71, p53 at Ser 15, c-Jun at Ser 73, and HSP27 at Ser 78. Total protein quantification was conducted using a detergent compatible protein assay kit (Bio-Rad, Hercules, CA, USA). Homogenized samples were diluted prior to being loaded, and results are reported as fluorescence intensity expressed relative to total protein content. To eliminate inter-assay variance, all tissue samples were thawed once and analyzed in duplicate in the same assay run by a single technician. The average coefficient of variation for phospho-protein analysis was 7.7 %.

Dietary logs

Participants were instructed to maintain their normal dietary intake leading up to experiment trials. Participants

were then instructed to record as accurately as possible everything they consumed during the 24 h prior to the first experimental trial. For the following experimental trial, participants were required to duplicate the content, quantity, and timing of their daily diet during the 24 h prior. Participants were instructed not to eat or drink (except water) within 10 h of reporting to the HPL for experimental trials.

Statistical analysis

Prior to statistical procedures, all data was assessed for normal distribution, homogeneity of variance, and sphericity. If the assumption of sphericity was violated, a Greenhouse-Geisser correction was applied. Signaling proteins were analyzed using a two factor (trial $\times$ time) analysis of variance (ANOVA) with repeated measures on time. In the event of a significant F ratio, LSD post-hoc tests were used for pairwise comparisons. For effect size, partial eta squared statistics were calculated, and according to Green et al. (2000), 0.01, 0.06, and 0.14 were interpreted as small, medium, and large effect sizes, respectively. Significance was accepted at an alpha level of $p \leq 0.05$, and all data are reported as mean $\pm$ SD.

Results

Resistance exercise protocol

Significant differences between trials were noted for workout volume ($p = 0.01$). Workout volume (sets $\times$ load $\times$ reps) was significantly greater for HV ($45,300 \pm 13,920$ kg) compared to HI ($33,634 \pm 5662$ kg).

MAPK signaling

No significance between trial interactions was noted for phosphorylation of MEK1 ($F = 0.03$; $p = 0.88$; $\eta^2 = 0.001$), ERK1/2 ($F = 0.01$; $p = 0.93$; $\eta^2 = 0.0001$), p38 ($F = 0.7$; $p = 0.41$; $\eta^2 = 0.04$), JNK ($F = 2.3$; $p = 0.15$; $\eta^2 = 0.11$), STAT1 ($F = 0.6$; $p = 0.44$; $\eta^2 = 0.03$), MSK1 ($F = 0.3$; $p = 0.59$; $\eta^2 = 0.02$), ATF2 ($F = 0.01$; $p = 0.93$; $\eta^2 = 0.0001$), p53 ($F = 0.02$; $p = 0.89$; $\eta^2 = 0.001$), c-Jun ($F = 0.1$; $p = 0.81$; $\eta^2 = 0.003$), or HSP27 ($F = 0.9$; $p = 0.36$; $\eta^2 = 0.05$). In addition, no significant differences over time were noted for phosphorylation of MEK1 ($F = 0.4$; $p = 0.52$; $\eta^2 = 0.02$), ERK1/2 ($F = 2.2$; $p = 0.15$; $\eta^2 = 0.11$), p38 ($F = 0.01$; $p = 0.72$; $\eta^2 = 0.01$), MSK1 ($F = 0.4$; $p = 0.53$; $\eta^2 = 0.02$), ATF2 ($F = 0.7$; $p = 0.41$; $\eta^2 = 0.04$), p53 ($F = 2.6$; $p = 0.13$; $\eta^2 = 0.13$), or c-Jun ($F = 0.1$; $p = 0.75$; $\eta^2 = 0.01$). However, significant time effects were observed for phosphorylation of JNK ($F = 0.181$; $p = 0.0001$; $\eta^2 = 0.50$), HSP27

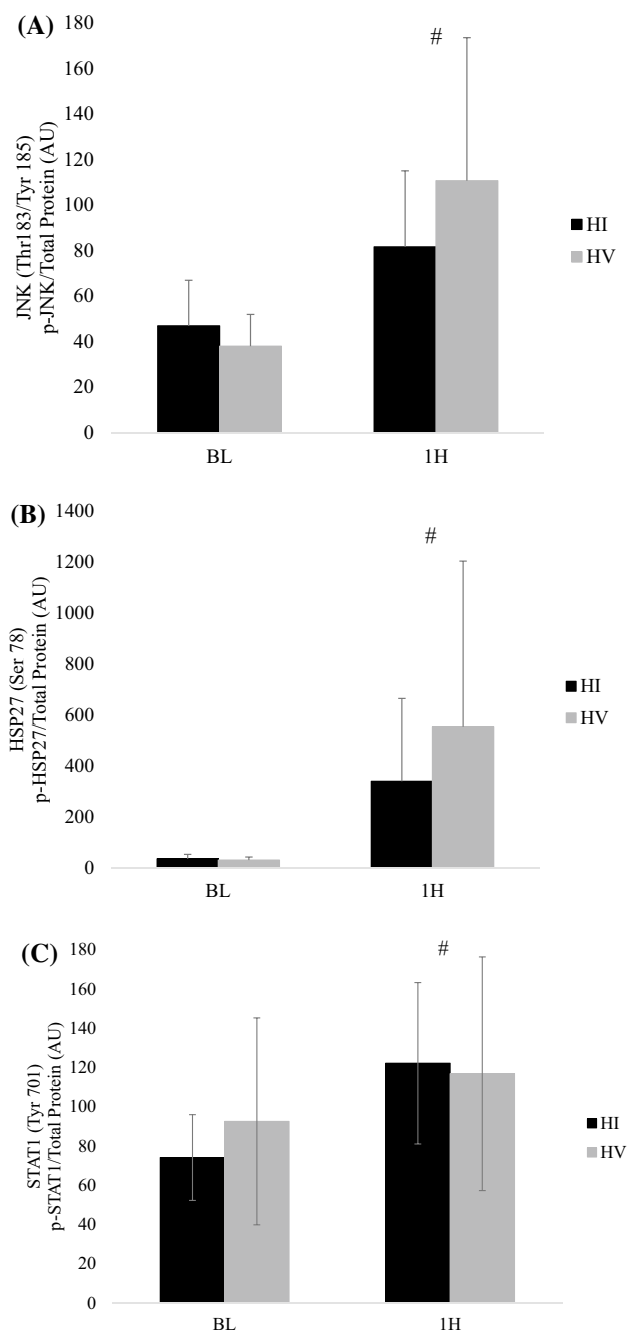


Fig. 1 Phosphorylation of **a** JNK (Thr 183/Tyr 185), **b** HSP27 (Ser 78), and **c** STAT1 (Tyr 701). Groups: *HI* high intensity resistance exercise protocol; *HV* high volume resistance exercise protocol. Time points: *BL* baseline; *1H* 1-h post. Phosphorylation status of signaling proteins was determined relative to total protein concentration and is, therefore, reported as arbitrary units (AU). All data are reported as mean $\pm$ SD. #Significant difference from BL ($p < 0.05$)

($F = 12.9$; $p = 0.002$; $\eta^2 = 0.42$), and STAT1 ($F = 5.8$; $p = 0.03$; $\eta^2 = 0.24$). Phosphorylation of JNK, HSP27, and STAT1 was significantly elevated from BL at 1H (Fig. 1).

Discussion

Resistance exercise is a potent stimulus for the remodeling of skeletal muscle (Egan and Zierath 2013). Among several intramuscular molecular networks, the MAPK signaling pathway appears to be a potential physiological mechanism involved in the exercise-induced regulation of gene expression in skeletal muscle (Widegren et al. 2001). However, the effect of different resistance exercise paradigms on early molecular signaling responses remains to be fully elucidated. The present study sought to examine MAPK signaling proteins following two different resistance exercise protocols in resistance-trained men. The HV and HI protocols were designed to be typical for a strength/power athlete focusing on hypertrophy and strength, respectively (Gonzalez 2016). The results of this study suggest that there is no differential response in the activation of MAPK signaling proteins when comparing HV and HI lower body resistance exercise protocols. However, both resistance exercise protocols resulted in significant elevations in JNK, HSP27, and STAT1 phosphorylation. No other signaling proteins were significantly augmented at 1H following resistance exercise.

Few studies have examined MAPK signaling following two distinct resistance exercise protocols. Among these investigations, MEK1, ERK1/2, and p38 appear to be the most commonly examined signaling proteins within the MAPK signaling pathway. Holm et al. (2010) reported greater ERK1/2 and p38 phosphorylation 30-min following work-matched unilateral knee extensions using 70 % 1-RM compared to 16 % 1-RM. However, Taylor and colleagues (2012) reported similar MEK1 and ERK1/2 activation following a leg press and leg extension protocol consisting of either 4 sets of 18–20 repetitions at 65 % 1-RM or 4 sets of 8–10 repetitions at 85 % 1-RM. Furthermore, Hulmi et al. (2012) reported greater ERK1/2 phosphorylation 30-min following a high-volume, moderate intensity leg press protocol (80 % 1-RM, 5×10 repetitions, 2-min rest interval) compared to a very low-volume, high-intensity leg press protocol (100 % 1-RM, 15×1 repetition, 3-min rest interval). However, Terzis et al. (2010) showed that MAPK signaling was not dependent on resistance exercise volume demonstrating that p38 phosphorylation was augmented to the same degree at 30-min following one, three, or five sets of 6-RM in the leg press exercise, while ERK1/2 phosphorylation remained unchanged following each trial. Thus, the effect of resistance exercise intensity and volume on the regulation of MAPK signaling remains unclear, as previous investigations have utilized a wide variation in intensity loads among previously untrained individuals. The major emphasis of our study was to examine the MAPK signaling response following HV and HI resistance exercise

paradigms using intensities greater than 70 % 1-RM. To the best of our knowledge, this appears to be the first study to compare MAPK signaling following distinct resistance exercise paradigms in resistance-trained individuals. Our findings suggest that the HV and HI resistance exercise protocols implemented in the current investigation elicit similar activation of MAPK signaling proteins in resistance-trained men.

While the HV and HI resistance exercise protocols implemented in the current study did not differentially augment MAPK signaling proteins, both resistance exercise protocols resulted in significant elevations in JNK, HSP27, and STAT1 phosphorylation at 1-H post-exercise. The lack of significant changes in other signaling proteins, including MEK1, ERK1/2 and p38 may be related to a lower adaptive ability among highly trained participants (Nader et al. 2014). JNK is a stress-activated protein kinase that leads to the activation of different transcription factors responsible for initiating gene expression and protein synthesis (Widegren et al. 2001). An increase in JNK phosphorylation has previously been observed following resistance exercise (Aronson et al. 1998; Boppart et al. 1999). STAT1 is activated by ERK1/2 and serves as a downstream nuclear transcription factors involved in muscle remodeling (Farooq and Zhou 2004; Guan 1994). An increase in STAT1 phosphorylation has been observed following a single bout of resistance exercise in rat skeletal muscle, and has been proposed to play a role in satellite cell proliferation (Begue et al. 2013). However, STAT1 phosphorylation has also shown to be unaltered following 3 sets of 12 repetitions of maximal single-leg knee extensions in young men (Trenerry et al. 2011). HSP27 appears to be activated by p38 and may be involved in processes important to cell repair following an exercise stimulus. HSP27 has also previously been shown to increase in response to a single bout of resistance exercise in human skeletal muscle (Thompson et al. 2003). Although minimal literature exists relating the expression of intramuscular HSP27 phosphorylation and resistance exercise, HSP27 has been suggested to serve as a molecular chaperone for intermediate filaments and may prevent stress-induced protein aggregation (Thompson et al. 2003). However, further research is needed to elucidate the role of heat shock proteins in the adaptive response to exercise.

The mechanisms that facilitate muscle protein synthesis following acute resistance exercise require the activation of signaling molecules within the mTORC1 (mammalian target of rapamycin complex 1) (Drummond et al. 2009; Kumar et al. 2009) and/or MAPK (Tannerstedt et al. 2009; Williamson et al. 2003) signaling cascades. Among the intramuscular signaling networks, mTORC1 appears to be the primary regulator of muscle protein synthesis and

growth (Goodman 2014). Although MAPK signaling may promote activation of mTORC1 signaling, MAPK signaling does not appear to be absolutely necessary for mechanically induced mTORC1 activation (Sasai et al. 2010; You et al. 2012). It remains unclear whether maximal stimulation of muscle protein synthesis is mediated by the convergence of mTORC1 and MAPK signaling pathways on downstream targets, such as p70S6 kinase (p70S6k) and ribosomal protein S6 (RPS6). Nevertheless, we recently demonstrated that the regulation of anabolic signaling proteins within mTORC1, including Akt, mTOR, p70S6k, and RPS6, was also similar following a lower body HV and HI resistance exercise protocol in resistance-trained men (Gonzalez et al. 2015b). Furthermore, in response to a single bout of resistance exercise, Kumar et al. (2009) demonstrated that p70S6k phosphorylation and muscle protein synthesis are dose dependent on intensity rising to a plateau at 60–90 % 1-RM, even when repetition number was increased at lower intensities to match work volume. Thus, it is plausible that an “intensity ceiling” of ~60 % 1-RM may exist, in which greater intensities would not confer further anabolism. Given that the HV and HI protocols implemented in the current study utilized 70 and 90 % 1-RM, respectively, this may, in part, explain the similar MAPK signaling response.

Investigating the effects of contractile intensity and volume on intramuscular anabolic signaling in vivo in humans certainly has inherent limitations. Although maximizing intramuscular anabolic signaling following resistance exercise has been suggested to produce the greatest potential for muscular adaptation with training, dissociation between anabolic signaling and muscle protein synthesis may exist (Greenhaff et al. 2008; Mitchell et al. 2015). Acute measures of muscle anabolism may not fully explain the dynamic process of muscle hypertrophy consequent to resistance training (Damas et al. 2015). In addition, the methods of studying intramuscular signaling in vivo provide a snapshot of the adaptive response at a single time point (i.e., 1H) following resistance exercise. Furthermore, the results of the current investigation are specific to highly trained individuals who may experience an attenuated signaling response in comparison to an untrained population (Nader et al. 2014).

In conclusion, the results of this study suggest that HV and HI lower body resistance exercise protocols elicit similar MAPK activation in resistance-trained men. Both resistance exercise protocols resulted in significant elevations in JNK, HSP27, and STAT1 phosphorylation, while MEK1, ERK1/2, p38, MSK1, ATF2, p53, and c-Jun were not altered by the exercise stimulus. Thus, it appears that emphasis on either training volume or intensity with loads greater than 70 % 1-RM may promote similar MAPK signaling responses following resistance training.

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